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# **Class 9 Biology Chapter 7**

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# Biology

## Unit no 7

### Metabolism

#### QUESTION NO 1: Define chemical reaction?

##### Answer: CHEMICAL REACTION

*Chemical reaction is a process in which one or more substances, the reactants, are converted to one or more different substances, the products is called chemical reaction*

#### QUESTION NO 2: Define biochemical reaction?

##### Answer: BIOCHEMICAL REACTION

*Those chemical reactions which are taking place in living organisms are called biochemical reactions.*

#### QUESTION NO 3: Define metabolism and its types?

##### Answer: METABOLISM

*The word metabolism means to change. The sum of all the chemical reactions going on within each cell of living organisms for the maintainance and sustainability of life is known as metabolism*

*There are 2 types of metabolism*

1 *Catabolic reaction*

2 *Anabolic reaction*

#### QUESTION NO 4: Differentiate b/w catabolic and anabolic reaction?

| <u>CATABOLIC REACTION</u>                                                                                                                                | <u>ANABOLIC REACTION</u>                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Biochemical reactions which break down complex molecules into simpler molecules, are called catabolic reactions</i>                                   | <i>Reactions which build complex molecules from simpler molecules are called anabolic reactions or anabolism.</i>                                                                              |
| <b>EXAMPLE</b><br><i>digestion is the catabolic process in which larger and complex food molecule are broken down into smaller and simpler molecules</i> | <b>EXAMPLE</b><br><i>photosynthesis is an anabolic process where simple molecule like <math>\text{CO}_2</math> and <math>\text{H}_2\text{O}</math> are converted to lager molecule glucose</i> |



**QUESTION NO 5: Define enzymes?**

**Answer: ENZYMES**

*Enzymes are biologically active globular proteins made by living cells, which tremendously speed up the biochemical reactions*

**QUESTION NO 6: Define substrate and product in short?**

**1: substrate:**

*Enzymes act on specific molecules in the cell called substrates.*

**2: product:**

*The substrates modified by enzymes are called products*

**QUESTION NO 7: Differentiate b/w intracellular and extracellular enzymes?**

**Answer:**

| <u>INTRACELLULAR ENZYMES</u>                                                                                | <u>EXTRACELLULAR ENZYMES</u>                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>The enzymes, which stay inside the cells to speed up the reactions, are called intracellular enzymes</i> | <i>The enzymes made inside the cells are allowed to go out of the cell to do their work outside. These enzymes are called extracellular enzymes</i> |
| <b><u>EXAMPLE</u></b>                                                                                       | <b><u>EXAMPLE</u></b>                                                                                                                               |
| <i>enzymes of glycolysis working in the cytoplasm</i>                                                       | <i>pepsin enzyme working in the stomach cavity</i>                                                                                                  |

**QUESTION NO 8: Why it is important to chew the food for a longer time. Explain?**

**Answer:**

*If you chew your food longer, more starch digesting enzymes amylase is produced and mixed to food that helps in breakdown of food especially starch. Moreover, the process of chewing also triggers the production of HCL in the stomach that regulate the lower pH in stomach to activate the enzyme pepsin, which breakdown the protein.*

**QUESTION NO 9: Define enzymology?**

**Answer: ENZYMOLOGY**

*Enzymology is the study of structure, types, function and role of enzymes is known as enzymology*

***Topic : 7.2.1***

***QUESTION NO 1 WHAT ARE THE Characteristics of enzyme?***



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### 1; enzyme are globular protein

- Enzyme are special type of protein
- They are made up of chains of amino acid that are folded into 3D complex shapes
- These shapes help them carry out their specific function
- Most enzyme are protien, but some are made up of RNA called ribosomes

### 2; Enzyme increase rate of reaction

- Chemical reaction in our body happen very slow without enzyme
- Enzyme make these reaction million of time faster
- This helps body to carry out important processes in seconds instead of months

### 3; Enzyme are required in small amount

- A very small amount of enzyme is enough to catalyse (speed up) a large amount of substrate
- Enzyme are not used up in the reaction-and remain unchage and can be used again

### 4; Enzyme are specific

- Each enzyme work on only one specific substrate
- For example amylase will only break down starch not protiens or Fats

### 5;Enzyme require co-factor

- Some enzyme need extra non-protien helper to work properly
- These helpers are called co-factors
- Co-factor help the enzyme bind to the substrate or complete the reaction
- Three types of co factor

1. **ACTIVATORS:** inorganic ions like ( $Zn^{2+}$ ,  $Fe^{2+}$ ) that help enzyme

2. **PROSTHETIC GROUP:** organic molecules that are tightly attached to the enzyme (e.g , FAD)

3. **COENZYME :** organic molecules thar are loosely attached ( e.g ,NAD , vitamins)

### 6; Regulation of Enzyme production and activity

- The body controls how many enzyme are made on base of its needs
- Enzyme activity can also be increased or decrease depending on the situation

### 7; Enzyme form metabolic pathway

- Enzyme work in step by step to carry out chain of reaction s
- This chain is called metabolic pathway



- Each step is done by a different enzyme, and product of one step become substrate For the next, until the final product is formed

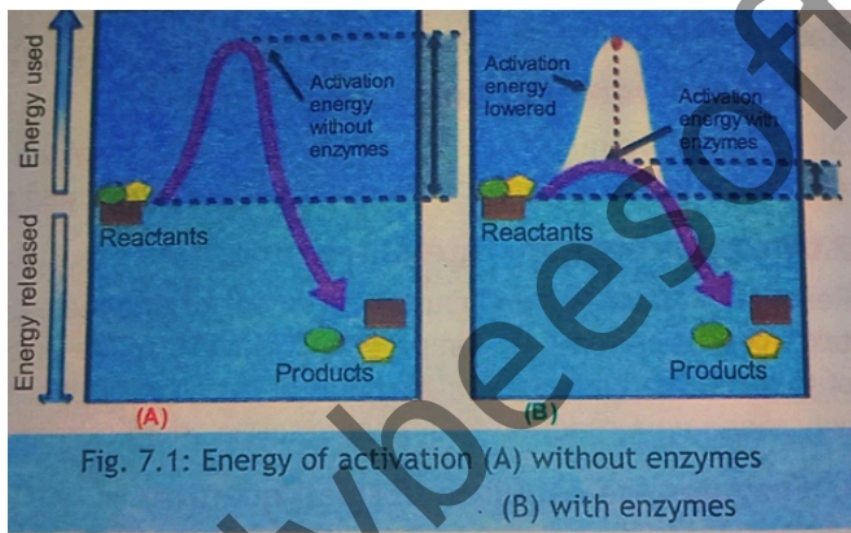
### 8;Energy of activation

- To start any Chemical reaction a small amount of energy is needed
- This energy is called activation energy
- Enzyme lower the activation energy, so the reaction can start more easily and quickly

### DIGRAM EXPLANATION

1 (A) without enzyme • High Peak ( high activation energy) • Reaction l'd slower

2 (B) with enzyme • Lower peak (lower activation energy)



### Topic :7.3

### Mechanism of enzyme action

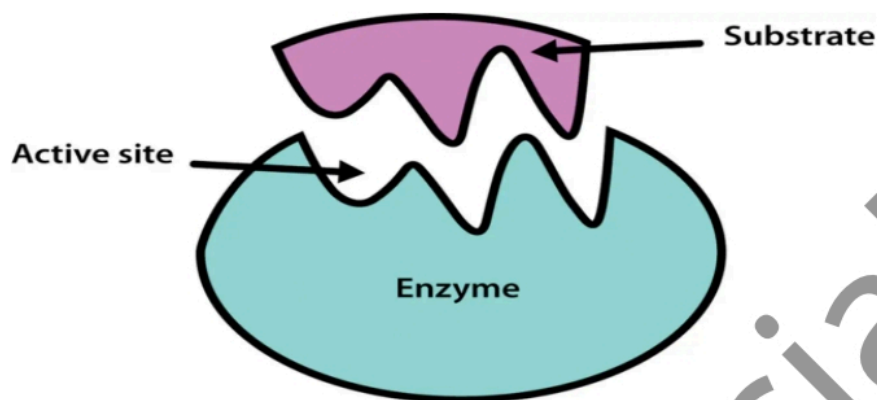
QUESTION NO 1: Define active site with the help of diagram?

Answer: Active Site:

The active site is the special part of an enzyme where the substrate attaches. It helps the enzyme to speed up the chemical reaction by fitting the substrate perfectly.



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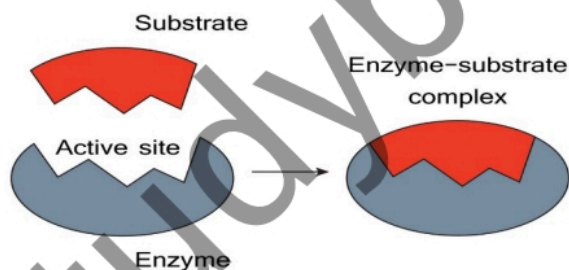
**QUESTION NO 2: Define Enzyme-Substrate Complex and enzyme-Product Complex?**

**Answer:**

**Enzyme-Substrate Complex:** *it is defined as "when substrate is attached to the active site of enzyme " it is known as Enzyme-Substrate Complex (ES).*

**Enzyme-product Complex:** *it is defined as "when substrate is converted into product "it is known as*

*Enzyme-product Complex (EP).*



**Question no 3: What are the two model that proposed the mechanism of enzyme?**

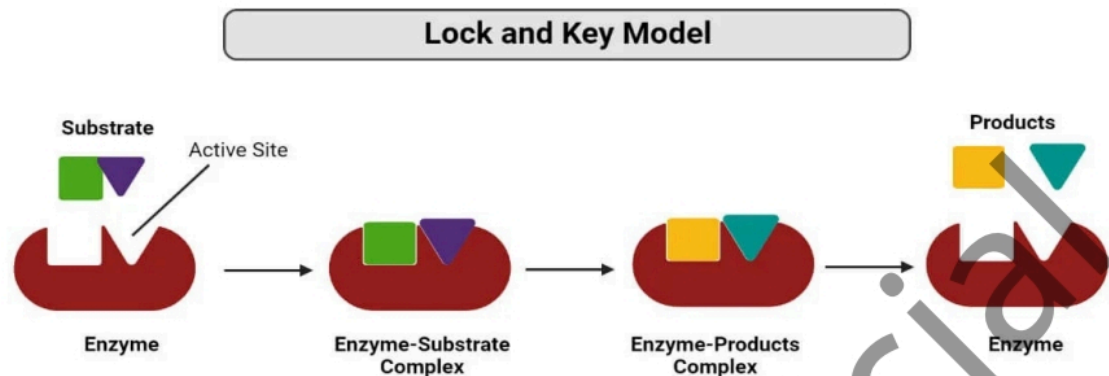
**Answer:** *There are 2 model that proposed the mechanism of enzyme are discuss below:*

**1: LOCK AND KEY MODEL:**

Lock and key model say that the active side of enzyme has a fixed structure. The substrate molecule fits precisely into it to form an "enzyme substrate complex" . The enzymes start the reaction and convert the substrate into the product "enzyme product complex". Then the product released from Enzyme .

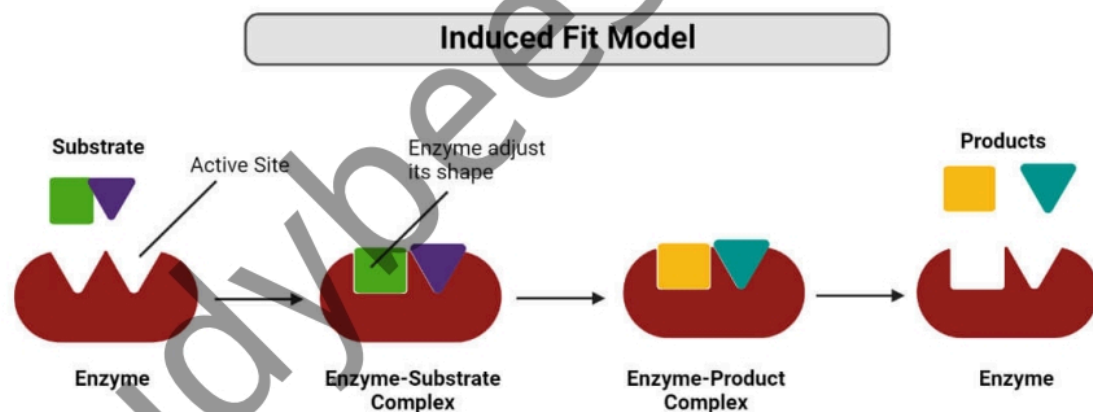


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## 2: INDUCED FIT MODEL:

*Induced fit model says that the active site of enzyme is not rigid. When substrate is not similar to the active site of enzyme the active site of enzyme reshaped itself according to substrate to perform its function.*



## TOPIC 7.4

### Factors affecting the enzyme activity

#### QUESTION NO 1: What are the factors that affects the enzyme activity?

Answer: *There are 3 factors that affects the enzyme activity are:*

1 *temperature*

2 *pH*

3 *substrate concentration*



### QUESTION NO 2: Define temperature with the help of diagram?

Answer: The rate of reaction increase with increase in temperature but up to specific limit.

#### Optimum Temperature:

Each enzyme work at the best temperature called **optimum temperature**. Different enzyme has different optimum temperature. Example. Optimum temperature for human enzyme is 36 centigrade to 38 centigrade. Any increase or decrease in temperature affect the rate of enzyme activity.

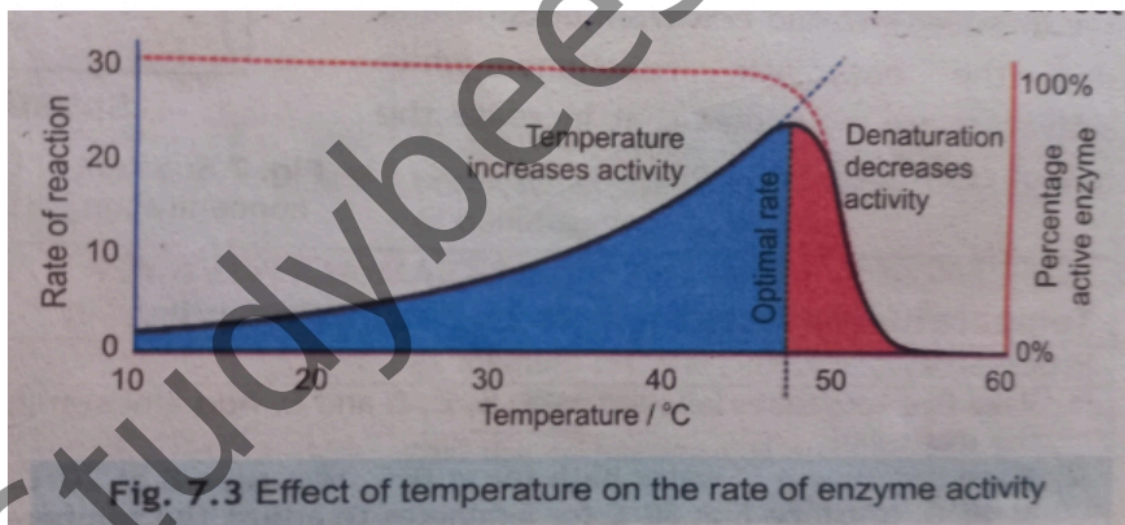
#### Effects of temperature on enzyme activity:

##### 1 high temperature:

At very high temperature atoms of the enzyme molecules start vibrating strongly so globular structure and active side of enzyme is damaged and it is called **denatured**

##### 2 low temperature:

At very low temperature enzyme remain inactive due to fall in available activation energy. They will regain the catalytic activity with normal temperature is restored.



### QUESTION NO 3: Define pH with the help of diagram?

Answer: pH:

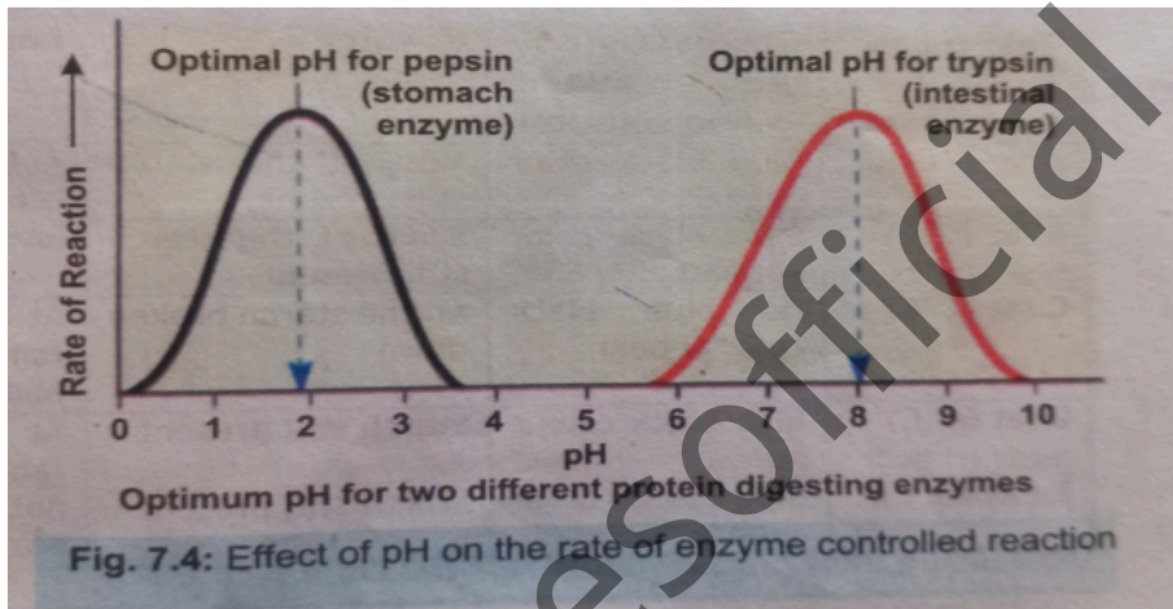
pH of medium affect the enzyme functioning. Each enzyme performed best at specific pH. Which is known as **optimum pH**. Slight change in pH of the reacting medium below or above the optimum pH reduce the rate of reaction. Extreme change in pH it denature the enzyme for example pepsin and trypsin are both protein digesting enzyme.

#### Pepsin and Trypsin Enzyme:



1 Pepsin enzyme performed best in acidic medium in stomach at 2.00 pH

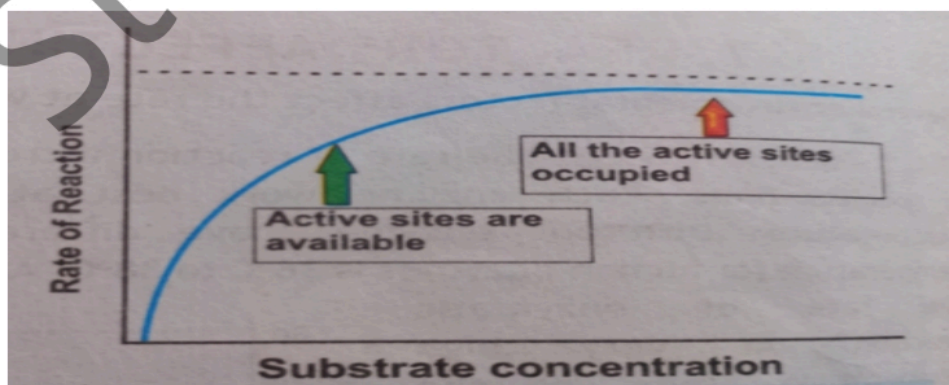
2 Trypsin enzyme performed best in alkaline medium in small intestine at 8.00 PH



QUESTION NO 4: Define substrate concentration with the help of diagram?

Answer: Substrate Concentration:

Substrate concentration refers to the amount of substrate available in a solution for enzymes to act upon in a biochemical reaction. The rate of enzymatic reaction increase with the increase in substrate Concentration. After that a point is reached when increase in substrate concentration will have no effect on the enzymatic reaction. It is due to the presence of extra substrate molecule which occupied all the active sites of available enzyme molecule this condition is known as **saturation of active site**.



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QUESTION NO 5: Define inhibitors and activators?

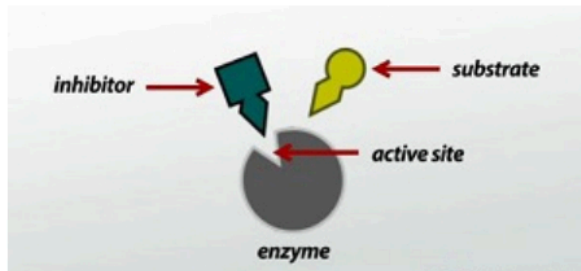
Answer:

1 inhibitors:

*Inhibitors are molecules produced in the cell or enter into the cell from outside and stop the enzymatic reactions.*

2 activators:

*Activators are molecules that increase the rate of enzymatic reaction.*



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Unit no. 7

Topic: Metabolism

Sub-Topic: Enzyme Inhibition

### 1. Define Enzyme Inhibition?

**Enzyme Inhibition:**

Enzymes are not able to perform their role due to the presence and interference of some chemicals at the reaction site, this phenomenon is called enzyme inhibition.

### 2. Define Inhibitor?

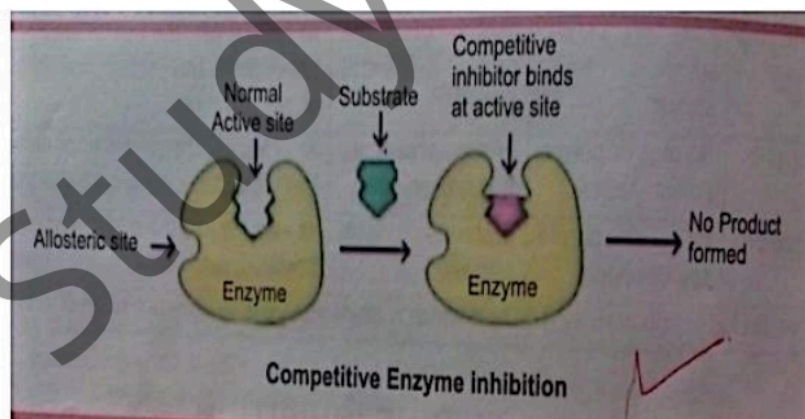
**Inhibitor:** The chemical substance which react with enzyme in place of substrate but does not convert to products thus inhibiting the enzyme action is called inhibitor.

### 3. Write the types of Enzyme Inhibition or Differentiate b/w Competitive & Non-competitive Inhibition?

**Competitive Inhibition:**

If enzymes are inhibited due to presence of molecules which occupy the active site due to their similarity in structure and shape to substrate is called competitive inhibition. Such inhibitors are neither converted to products nor they allow the actual substrate to bind.

Competitive Inhibition is temporary and can be reversed if the concentration of actual substrate is increased.



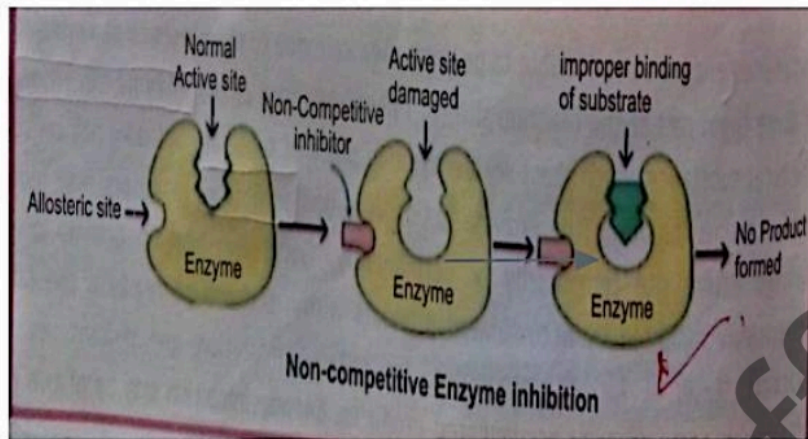
**Non-competitive Inhibition:**

The structure and shape of these inhibitors do not resemble the substrate molecule so they do not occupy the active site. They may attach to enzyme surface at a point other than active site



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called allosteric site. They alter the globular shape of enzyme and damage the active site of enzyme, Thus, actual substrate will not be able to bind to active site and reaction stops.



## Topic 7.6 Role of Atp

### 1. Describe the structure and uses of ATP?

#### ATP

The living organisms use and store energy at the cellular level in the form of Adenosine triphosphate (ATP). ATP is commonly called the "energy currency" of cell.

#### Structure of ATP:

ATP molecule consists of three components:

- Adenine (double ringed Nitrogen Base)
- Ribose (Five Carbon Sugar)
- Three phosphate Groups ( $\text{PO}_4$ ) linked in a chain called a triphosphate group

Adenine nitrogen base binds to Ribose sugar and become adenosine.



**AMP:** First phosphate group binds with adenosine to form adenosine monophosphate (AMP).

**ADP:** Second phosphate group binds with AMP to form adenosine diphosphate (ADP).

**ATP:** Third phosphate group binds with ADP to form adenosine triphosphate (ATP).

Thus all three phosphate group are serially bonded in a linear chain with adenosine to form ATP.

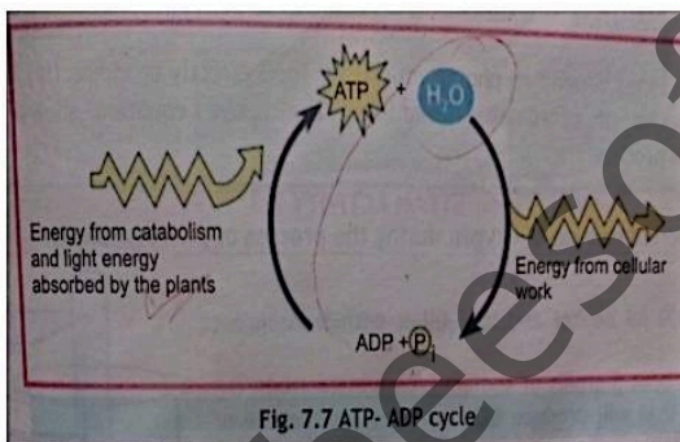


### Uses of ATP:

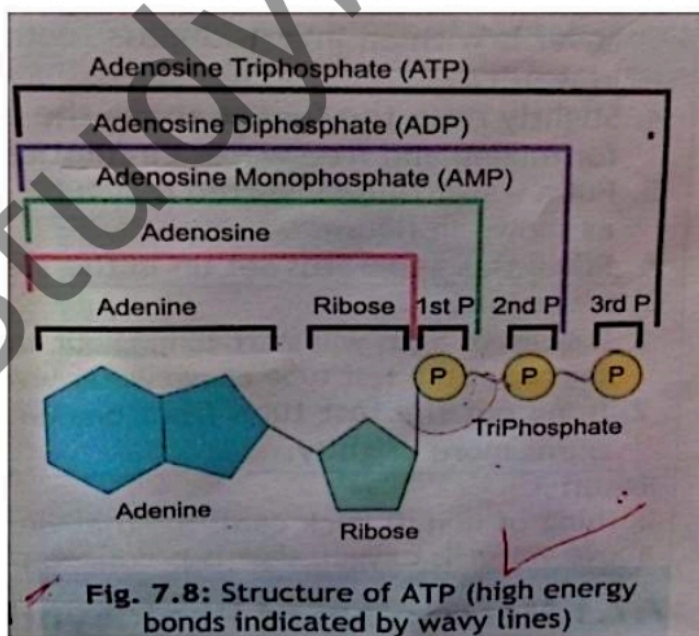
ATP is used as an energy source for a variety of cellular processes, including muscle contraction, ion transport across membranes, and the synthesis of proteins and lipids. It is also used for energy-intensive processes such as cell division and growth.

### 2. Explain briefly ATP-ADP Cycle?

ATP-ADP Cycle: ATP can be converted to ADP and inorganic phosphate by hydrolysis. The third phosphate group separates from ATP and remains in the cell in inorganic form. ADP and phosphate can be converted back to ATP by condensation. The ability of ATP to store and release energy is because of its unique structure.



### 3. Sketch the structure of ATP?



## Topic 7.7 Photosynthesis

### 1. Describe and sketch light reactions of photosynthesis.

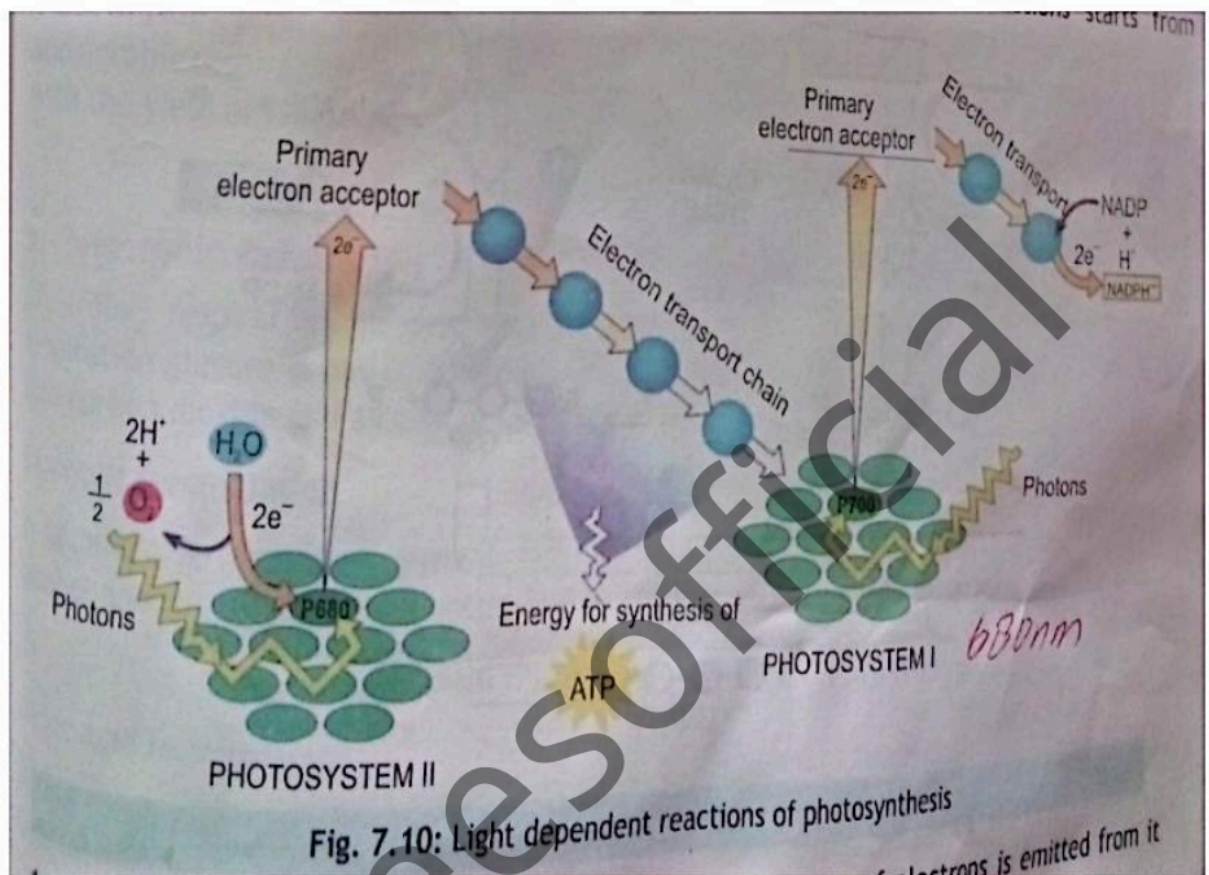
**Photosynthesis:** During the process of photosynthesis plants absorb light energy in the presence of chlorophyll and use carbon dioxide and water to make glucose.

#### **Mechanism of light Dependent Reactions of photosynthesis:**

The reactions, which depend upon light energy, are called light dependent reactions of photosynthesis. These reaction take place in thylakoid membrane where photosynthesis pigments are arranged into clusters called photosystems. There are two types photosystems: Photosystems I and photosystem II. Process of light dependent reactions starts from photosystem II.

1. Chlorophyll a of Photosystem II absorbs light and a pair of electrons is emitted from it which are accepted by a molecule called primary electron acceptor.
2. At the same time, water molecule also splits into oxygen atom, two hydrogen ions ( $H^+$ ) and two electrons. Splitting of water takes place to compensate the electron loss of light affected chlorophyll of photosystem II and is called photolysis of water.
3. Oxygen is released out of leaf through stomata.
4. Electron pair from primary electron acceptor passed through a chain of molecules called electron transport chain (ETC). Electron passing through ETP, gradually release energy which is used for ATP synthesis.
5. Light also acts on photosystem I which also gives out an electron pair. These electrons and two  $H^+$  of water reduce NADPH.





## 2. Describe and sketch dark reactions of photosynthesis.

**Light independent reactions:** During the dark reactions of photosynthesis, carbon dioxide is reduced by NADPH to form glucose molecule. This reduction requires energy which is provided by ATP. These reactions do not use light directly; hence they are known as light independent reactions. It takes place in stroma of chloroplast.

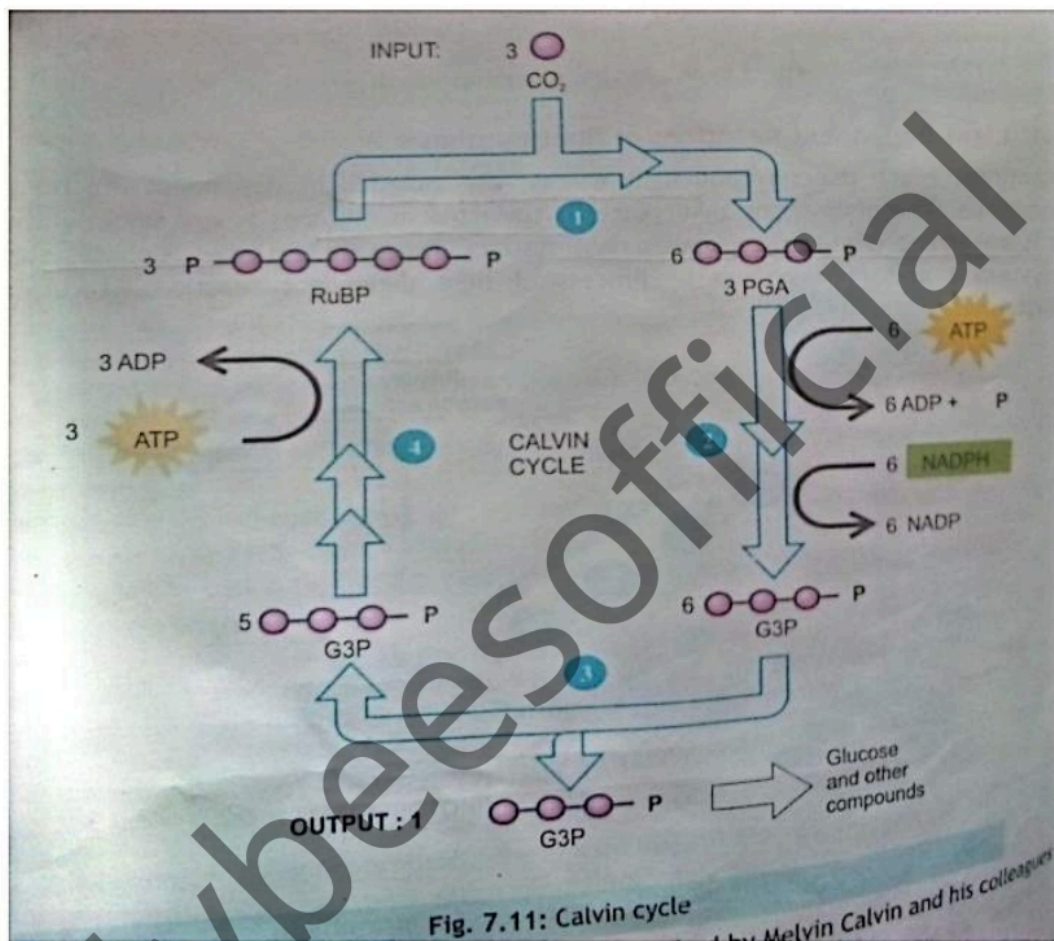
**Calvin Cycle:** The sequence of reactions in this phase was first described by Melvin Calvin and his colleagues so is known as Calvin Cycle as well.

1. Carbon dioxide combines with an already existing 5-carbon molecule (RuBP) to form a 6-carbon intermediate molecule.
2. This 6-carbon molecule is unstable and splits into two 3-carbon molecules (3PGA).
3. Now these 3-carbon molecules are reduced to 3-carbon sugar molecule (G3P) by NADPH using energy of ATP.



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4. These 3-carbon sugar molecules are used to synthesize glucose and also to regenerate original 5-carbon molecule for next Calvin Cycle.



3. Explain the factors affecting rate of photosynthesis.

Limiting Factors of Photosynthesis:

Following are the limiting Factors which affect at the rate of of photosynthesis:

- More light intensity and duration of light increase the process of photosynthesis.
- More CO<sub>2</sub> concentration increase the rate of photosynthesis but only up to a specific point.
- Rate of photosynthesis is highest at 25°C called optimum temperature for photosynthesis. Temperature below or above the optimum affect the rate of photosynthesis.

4. Write two phases of photosynthesis?



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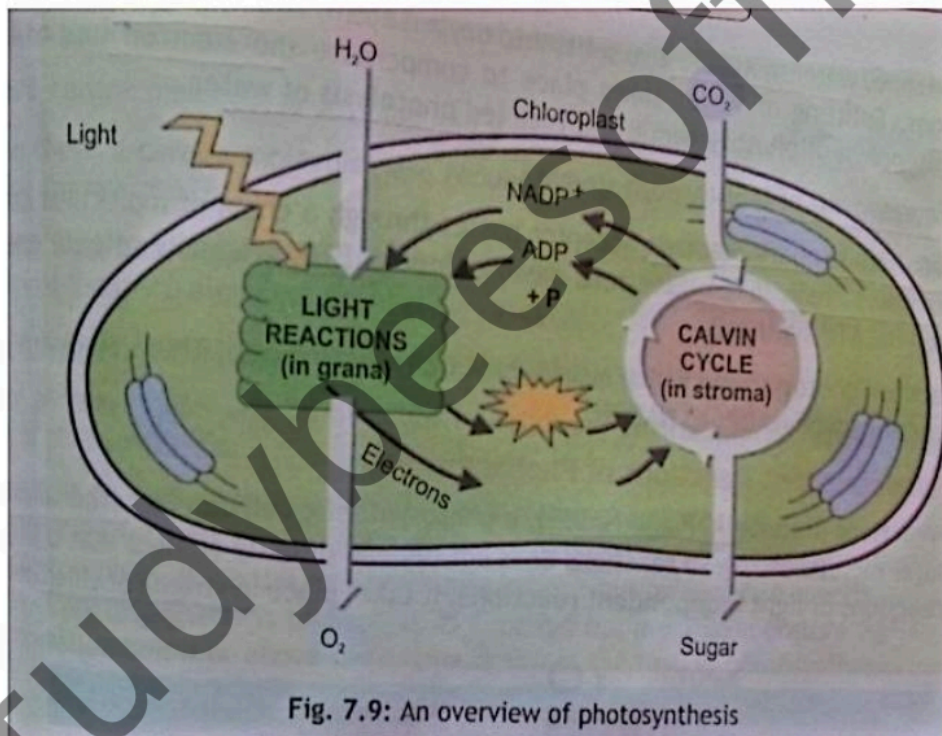
### Mechanism/Phases of Photosynthesis:

Photosynthesis occur in two phases

- Light dependent reactions
- Light independent reactions

**Light dependent reactions:** In the first phase, light energy is absorbed and then used to produce energy rich molecules ATP and NADPH, These reactions take place in chloroplasts.

**Light independent reactions:** During the second phase carbon dioxide is reduced by NADPH to form glucose molecule. This reduction requires energy which is provided by ATP. These reaction do not use light directly; hence they are known as light independent reactions.



### Topic 7.8: Respiration

#### 1. Define cellular Respiration?

**Respiration:** The process in which energy is produced in the form of ATP by the oxidation of glucose is called respiration. Respiration is a chemical process which takes place in cells.

#### 2. Differentiate b/w Anaerobic respiration and aerobic respiration?

| Anaerobic Respiration | Aerobic Respiration |
|-----------------------|---------------------|
|-----------------------|---------------------|



|                                                                                                                                                              |                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anaerobic Respiration means, respiration in the absence of oxygen.                                                                                           | Aerobic respiration means, respiration in the presence and use of oxygen.                                                                                                              |
| During Anaerobic respiration glucose is not completely oxidized to carbon dioxide and water but is converted into carbon dioxide and alcohol or lactic acid. | During aerobic respiration, glucose molecule is completely broken down in the presence of oxygen, to produce large amount of energy along with production of carbon dioxide and water. |

## 2. Define Fermentation?

**Fermentation:** During Anaerobic respiration glucose is not completely oxidized to carbon dioxide and water but is converted into carbon dioxide and alcohol or lactic acid. This process is called fermentation.

## 3. Write the two types of fermentation?

**Types of fermentation:** There are two types of fermentation.

- Alcoholic fermentation
- Lactic acid fermentation

**Alcoholic fermentation:** In alcoholic fermentation, pyruvic acid produced during glycolysis is converted into ethyl alcohol and carbon dioxide. It occurs in yeast and some bacteria.



**Lactic acid fermentation:** During lactic acid fermentation, each pyruvic acid molecule is converted into lactic acid. Bacteria which convert milk to yogurt also produce lactic acid.



## 4. Define Glycolysis?

During glycolysis, glucose is oxidized into two molecules of pyruvic acid with the yield of two ATP molecules.



## 5. Write the chemical equation of Aerobic Respiration?

**Aerobic Respiration:** During aerobic respiration, glucose molecule is completely broken down in the presence of oxygen, to produce large amount of energy along with production of carbon dioxide and water.





**6. What are the advantages and significance of anaerobic respiration and fermentation in your daily life?**

**Importance of Anaerobic Respiration:** Anaerobic Respiration is important for living organisms in following ways:

- The earliest living organisms would have to produce energy by anaerobic respiration because the early earth had no free oxygen in its atmosphere.
- Anaerobic Respiration by bacteria is used in cheese and yogurt making.
- Fermentation by yeast is used in wine making and baking industry.
- Anaerobic Respiration, in the form of lactic acid fermentation provides some energy to muscle cells during running, labour or strenuous exercise.

**7. Write the mechanism of Aerobic Respiration?**

**Mechanism of Aerobic Respiration:** The complete process of aerobic respiration is divided into four main phases:

- Glycolysis
- Formation of Acetyl CoA
- Krebs Cycle
- Electron Transport Chain

### 1. Glycolysis

Glycolysis takes place in cytoplasm, outside the mitochondria. It occurs both in anaerobic and aerobic respiration. A single molecule of glucose is broken down into two molecules of pyruvic acid having 3 carbons. Two ATP molecules are produced as net energy gain. Two  $\text{NAD}^+$  molecules are reduced to NADH.



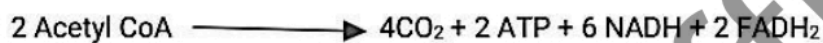
### 2. Formation of Acetyl CoA

Each pyruvic acid molecule is now oxidized to a two carbon Acetyl group which also combines with coenzyme A to form acetyl CoA. Carbon dioxide is removed and NADH is produced. The acetyl CoA enters mitochondria where it will take part in kreb Cycle.



### 3. Kreb Cycle

Kreb Cycle is a cyclic process which takes place in mitochondria matrix. Coenzyme A is released and acetyl group is passed through a series of reactions. The products of this process are  $\text{CO}_2$ , NADH and  $\text{FADH}_2$ . Some energy is released to produce ATP directly, while more energy is produced by Oxidation of NADH and  $\text{FADH}_2$  in ETC.



### 4. Electron Transport Chain (ETC)

Electron Transport Chain is a series of electrons carrier molecules located in the inner membrane of mitochondria. The electrons that are removed by Oxidation of glucose molecules during glycolysis and kreb Cycle are carried to ETC by NADH and  $\text{FADH}_2$ .

When electrons coming through NADH pass through ETC, they provide enough energy to synthesize three ATP molecules. However, each  $\text{FADH}_2$  produces only two ATP molecules. The final electron acceptor is oxygen atom which combines with hydrogen ions to produce water.

